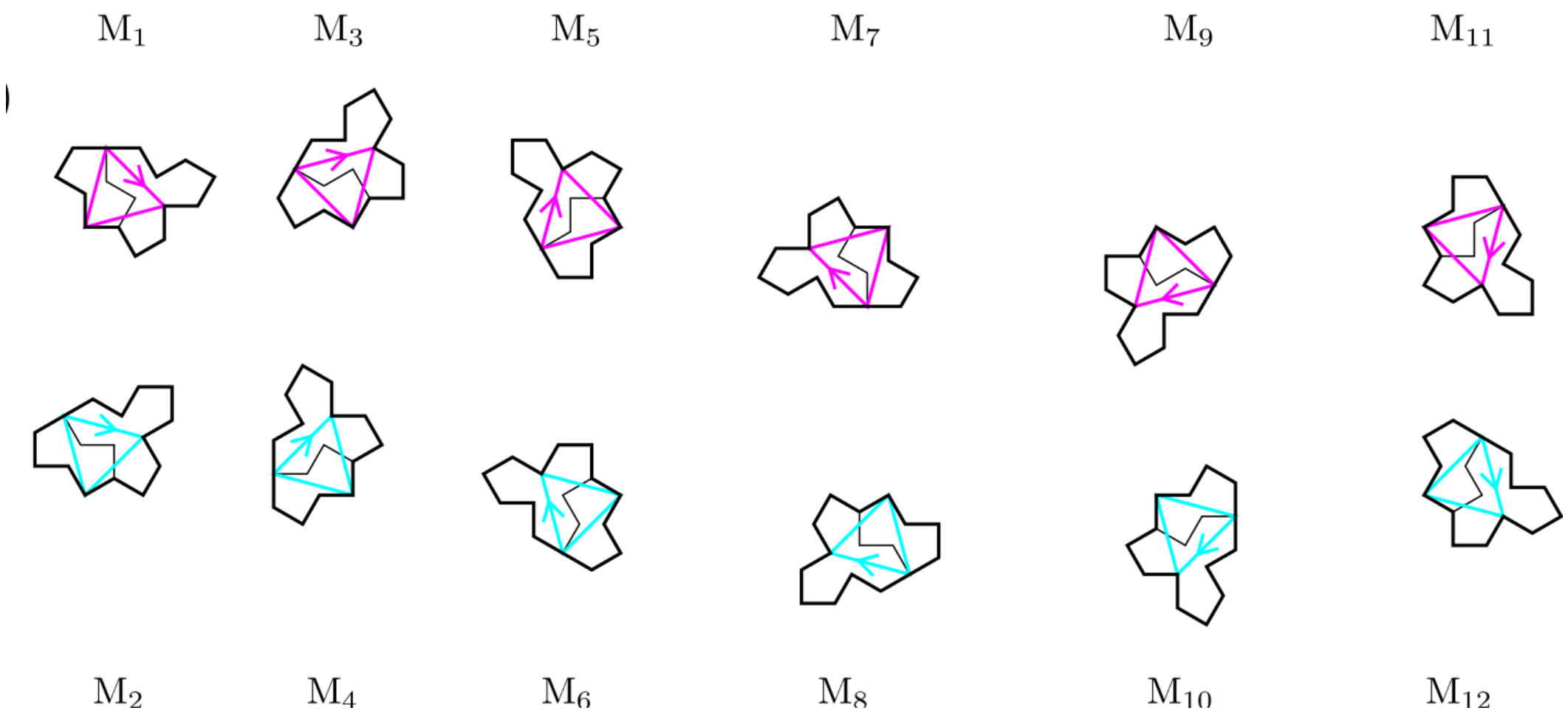
Homochiral ideas

Marianne Imperor-Clerc

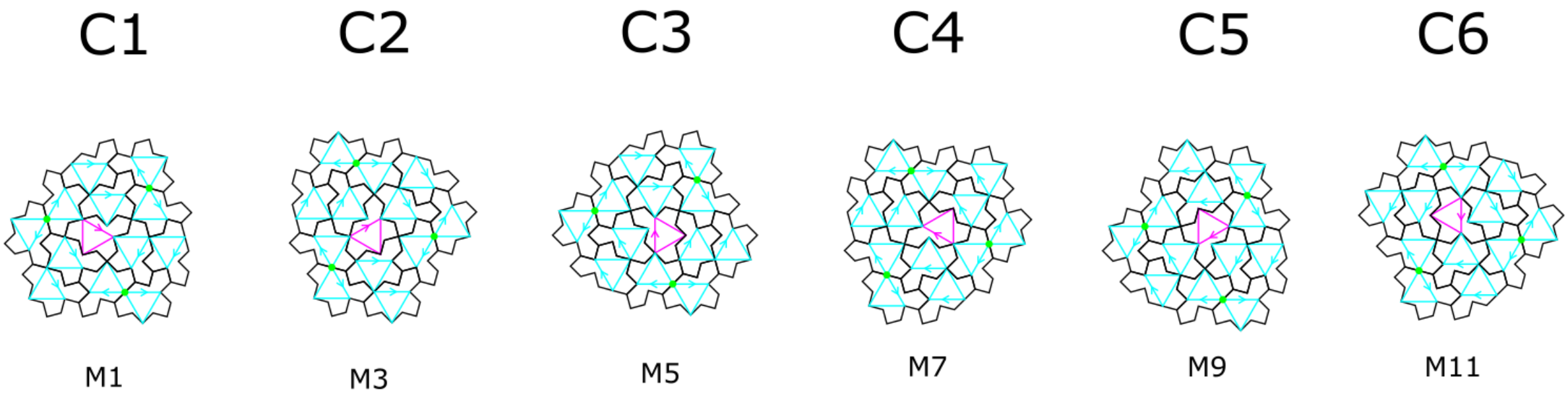
October 2025

ODD and EVEN orientations for Tile(1,1)



full cluster with its six rotations: n(2pi/6)

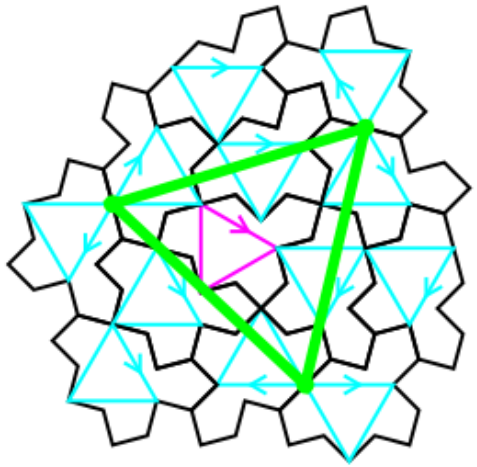
- Tile(1,1)



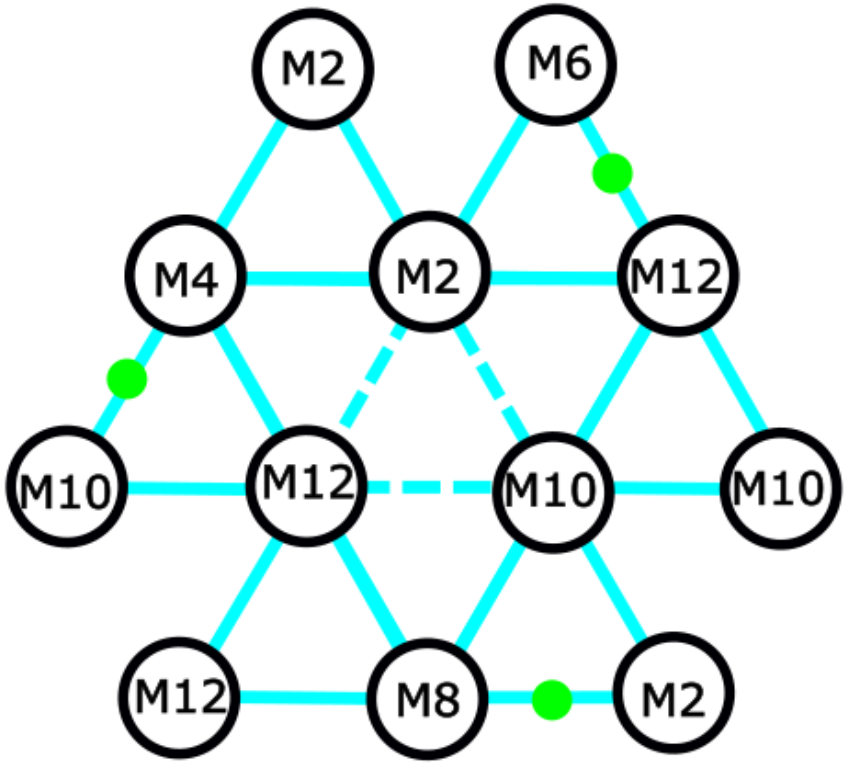
- 13 tiles in a full cluster: one ODD tile at the center + 12 EVEN tiles around
- Blue triangles (EVEN) have one horizontal edge when pink (ODD) have a vertical one

Orientation of the 12 EVEN tiles in a full cluster

- Tile(1,1)



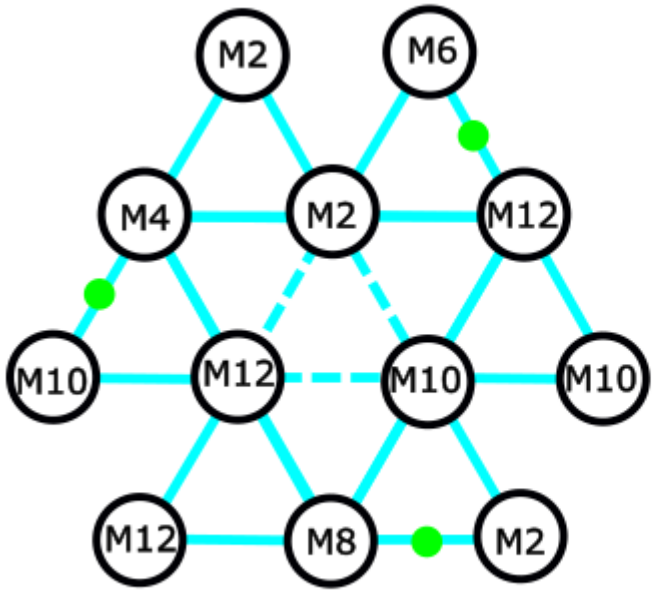
C1



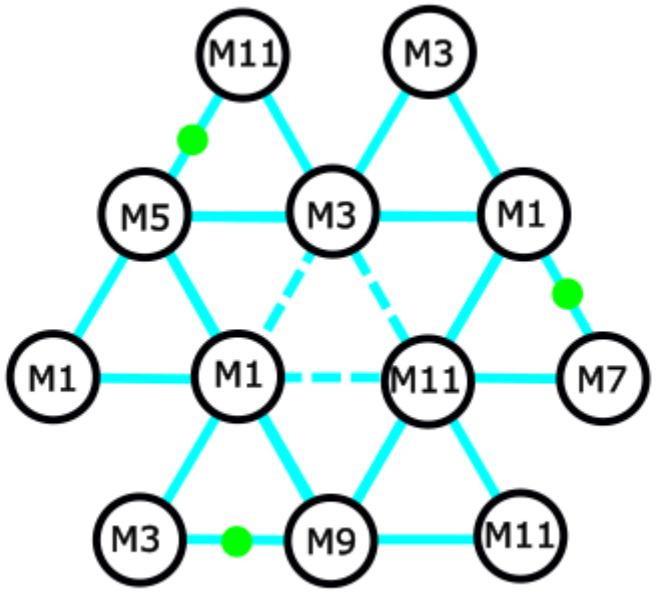
- 6 tiles in the C-cluster (corona made of M12, M4, M2, M12, M10, M8)
- 3 translated (M2, M10, M12) (along blue lines, the edges of blue triangles)
- 3 half-turned (M4->M10, M12 -> M6, M8 -> M2)

Inflation step 1: substitution rule for C1

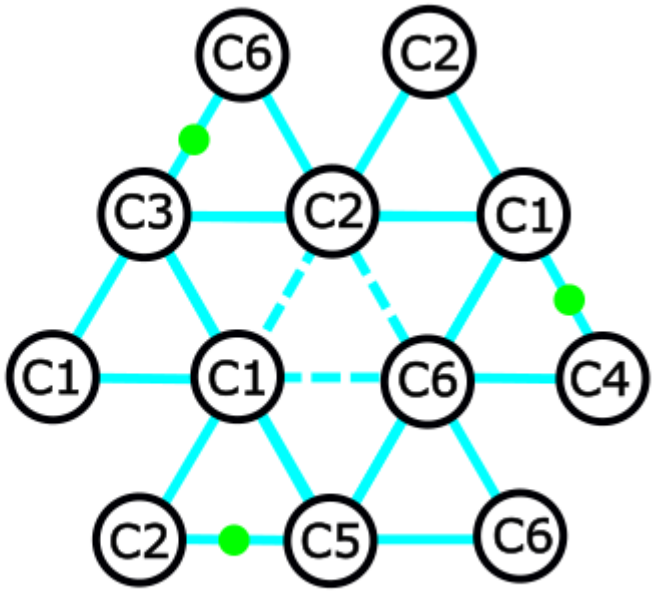
EVEN tiles in C1



ODD tiles for step 1

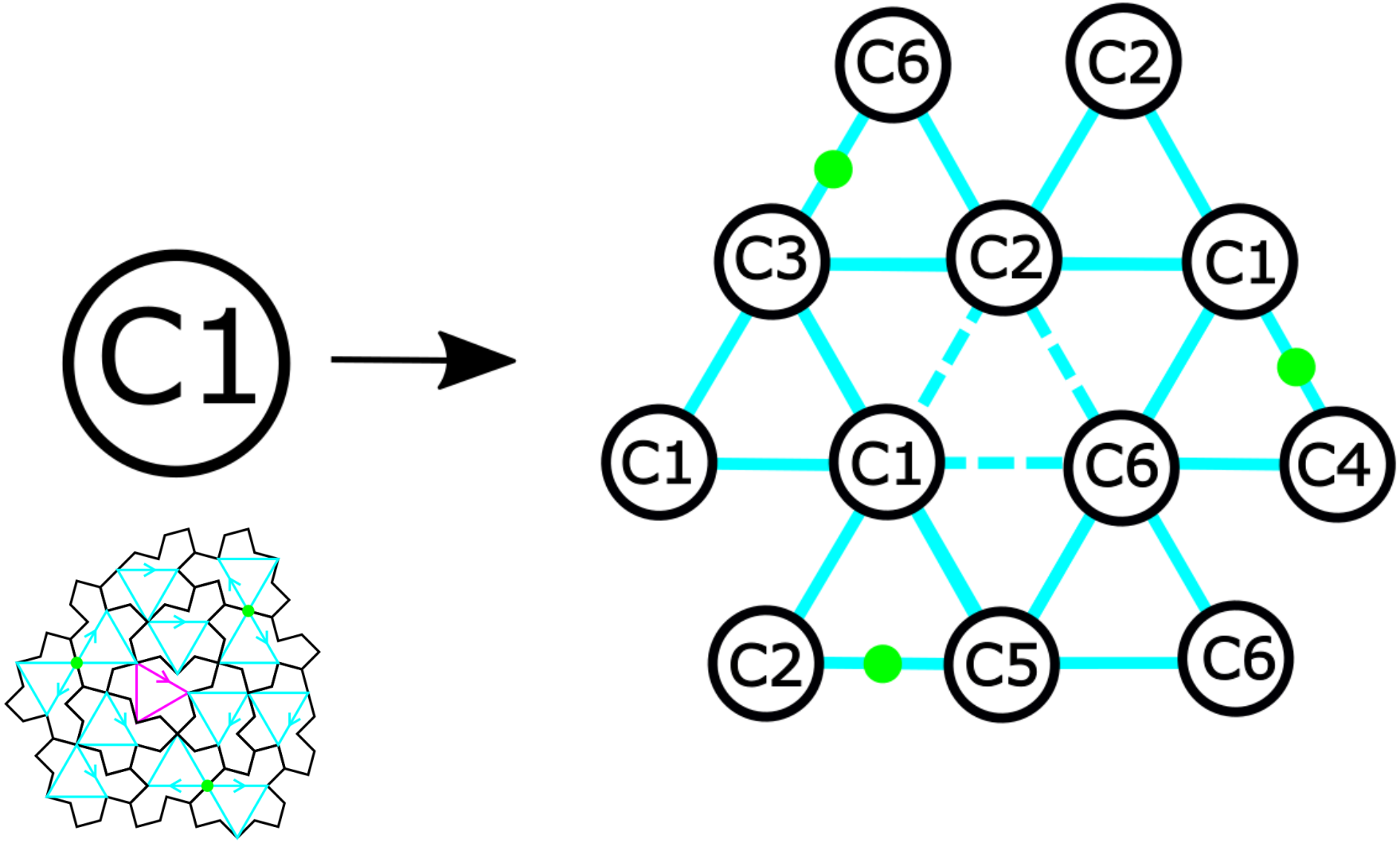


same as full clusters

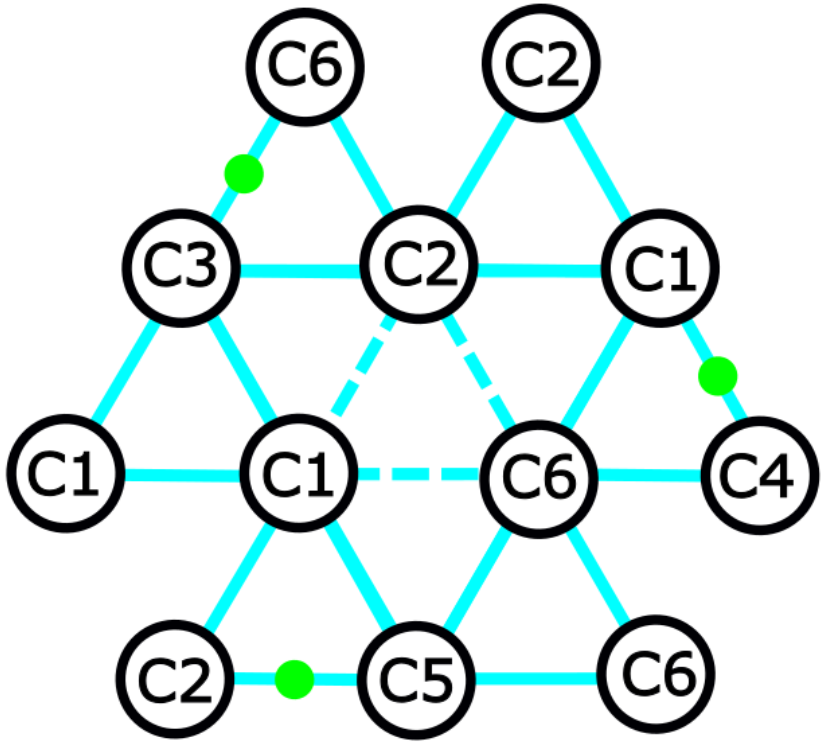


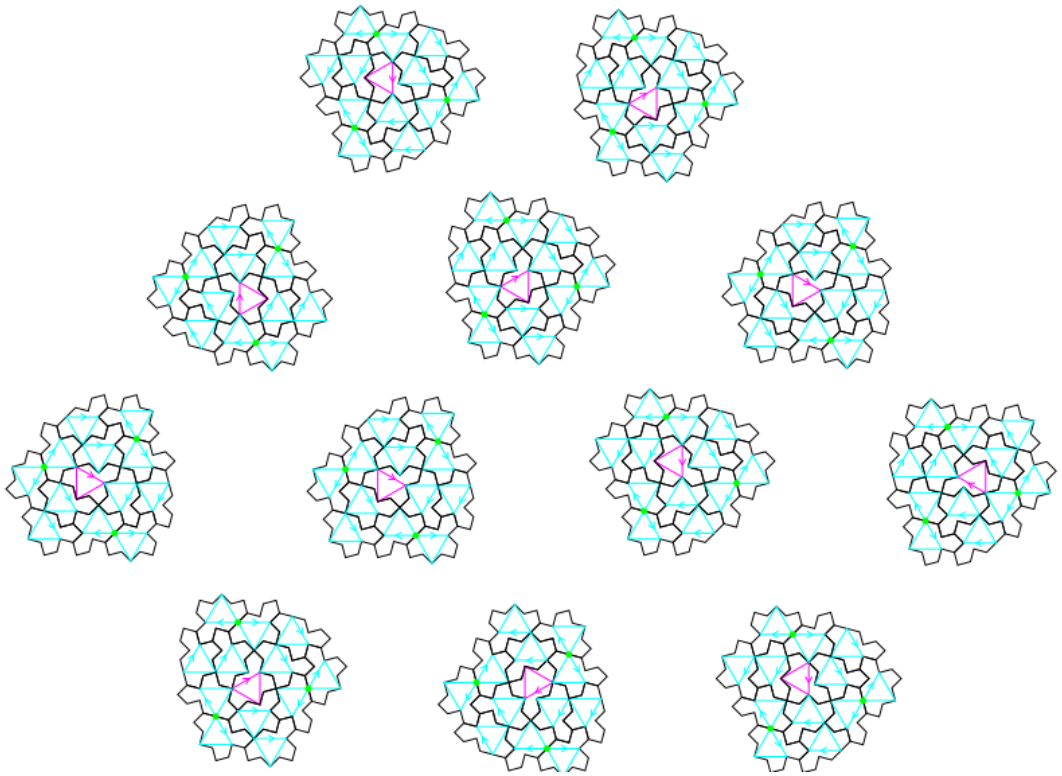
- ODD tiles for step 1 are obtained by adding 1 (mod 12) to all EVEN tiles in the C cluster (6 tiles)
- Directions of translated and half-turned tiles are exchanged (**see the change of position for green points**). Then three tiles are translated (M1, M3, M11) and three are half-turned (M5->M11, M1 -> M7, M9 -> M3)
- ODD tile index is equivalent to the orientation of a full C cluster ☺

Inflation step 1



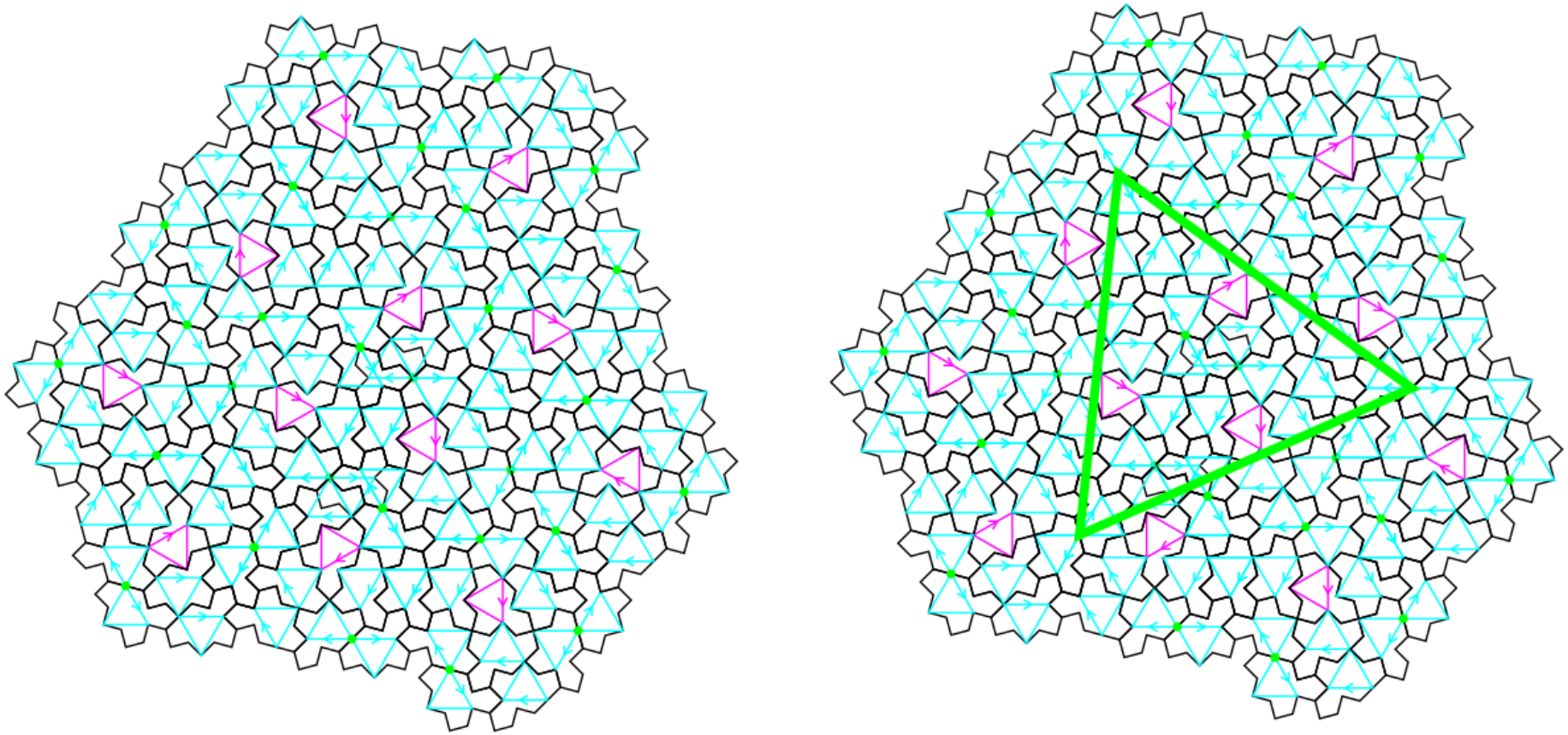
Inflation step 1





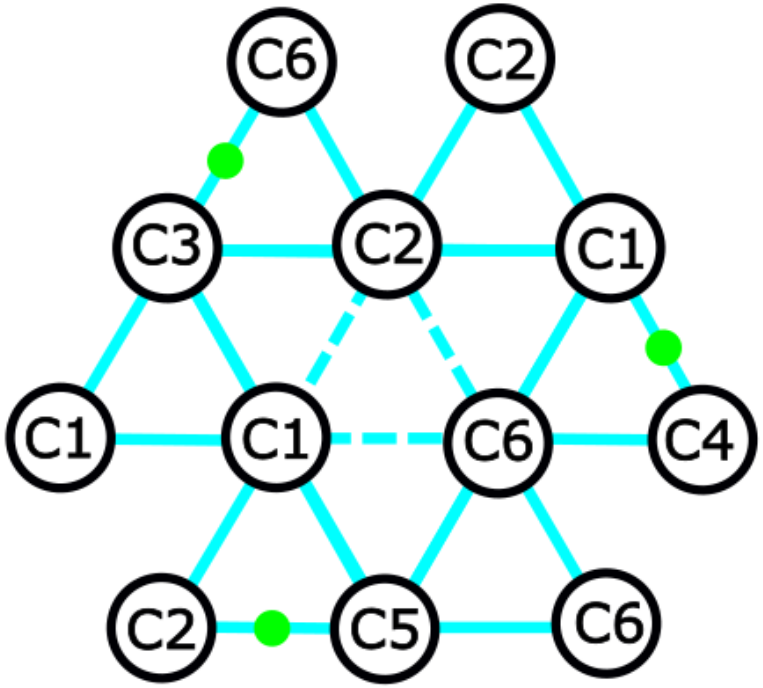
- Directions along the blue lines are the edge directions of the blue triangles !!!

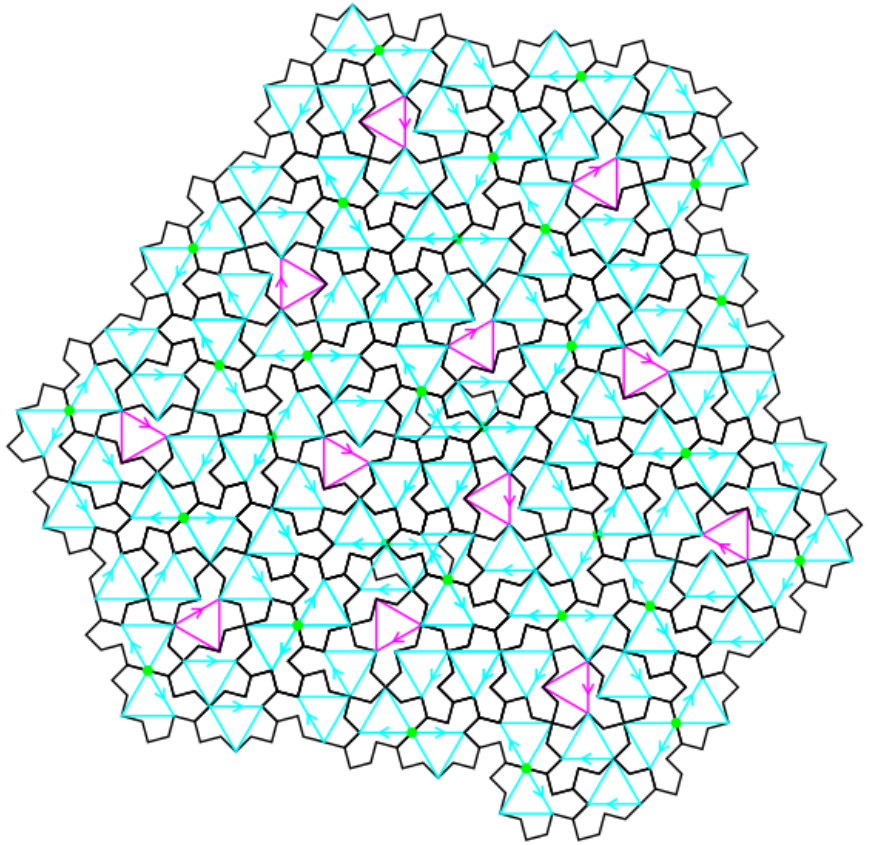
Inflation step 1 for full cluster C1



- Assembly of the 12 full clusters works fine !!! (done manually so far for the relative positions of clusters)
- Some tiles are superimposed
- Some half-turned tiles need to be removed with a simple rule: **translation wins over half-turn** !
- A bigger green triangle is identified, centered on a trifle at the center of the figure ☺

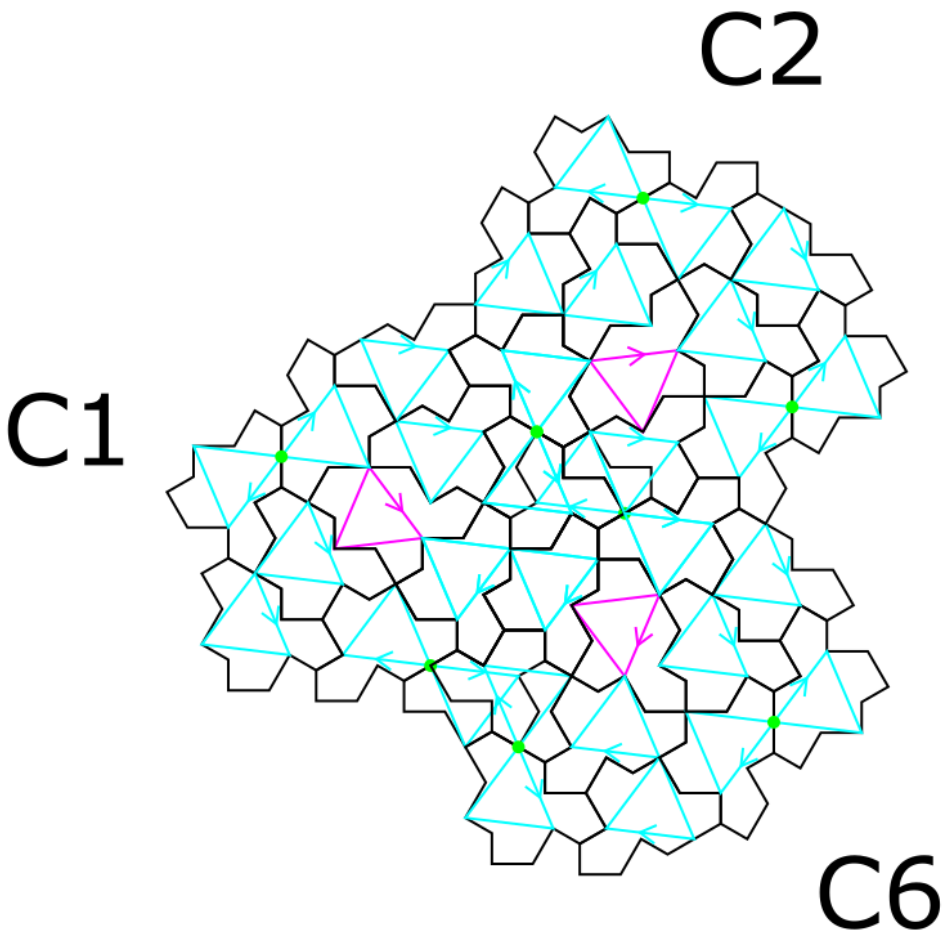
Inflation step 1 for full cluster C1

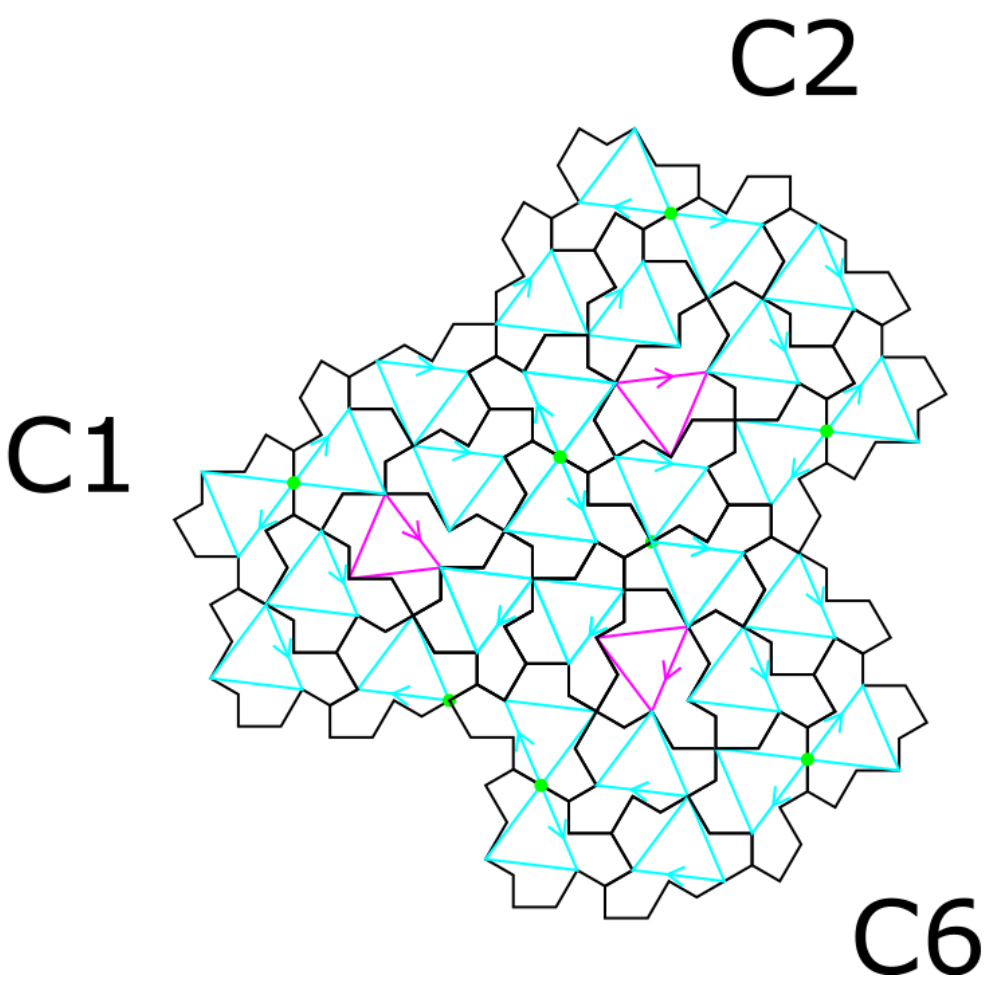




- TO DO: Find a rule for the relative positions between clusters in the assembly
- Clusters are translated from each other along the blue lines
- Use superposition of green points from two adjacent clusters ???

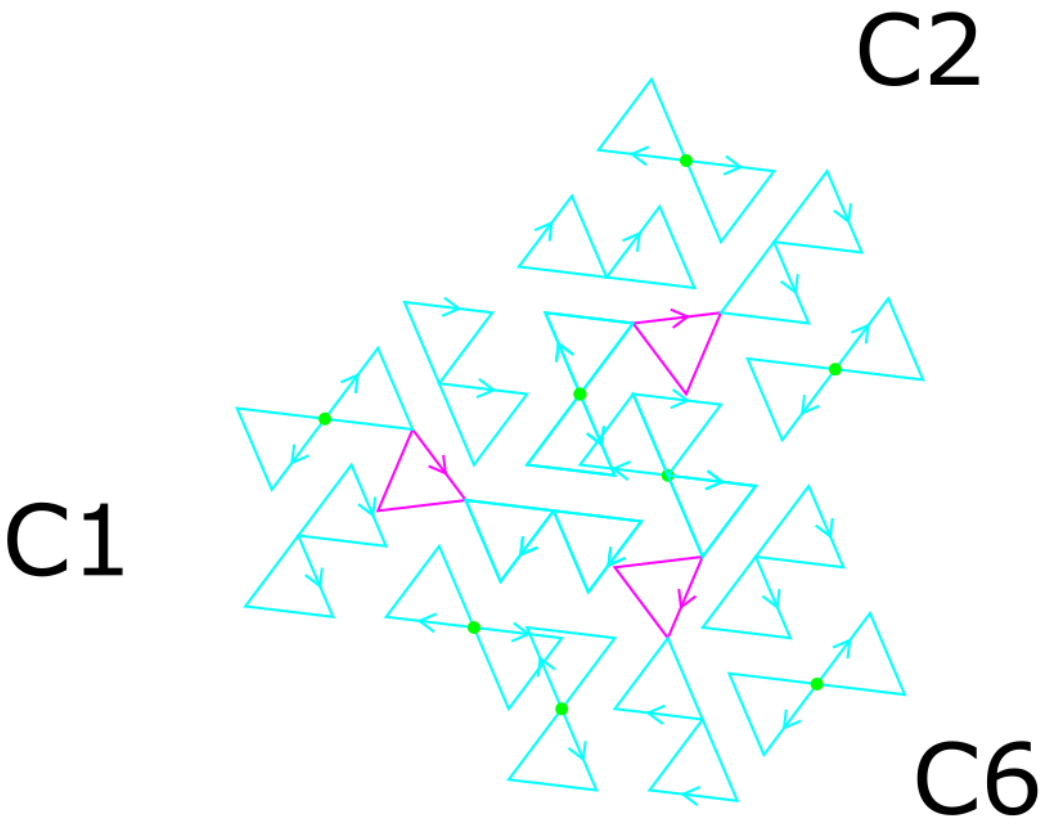
Defect healing

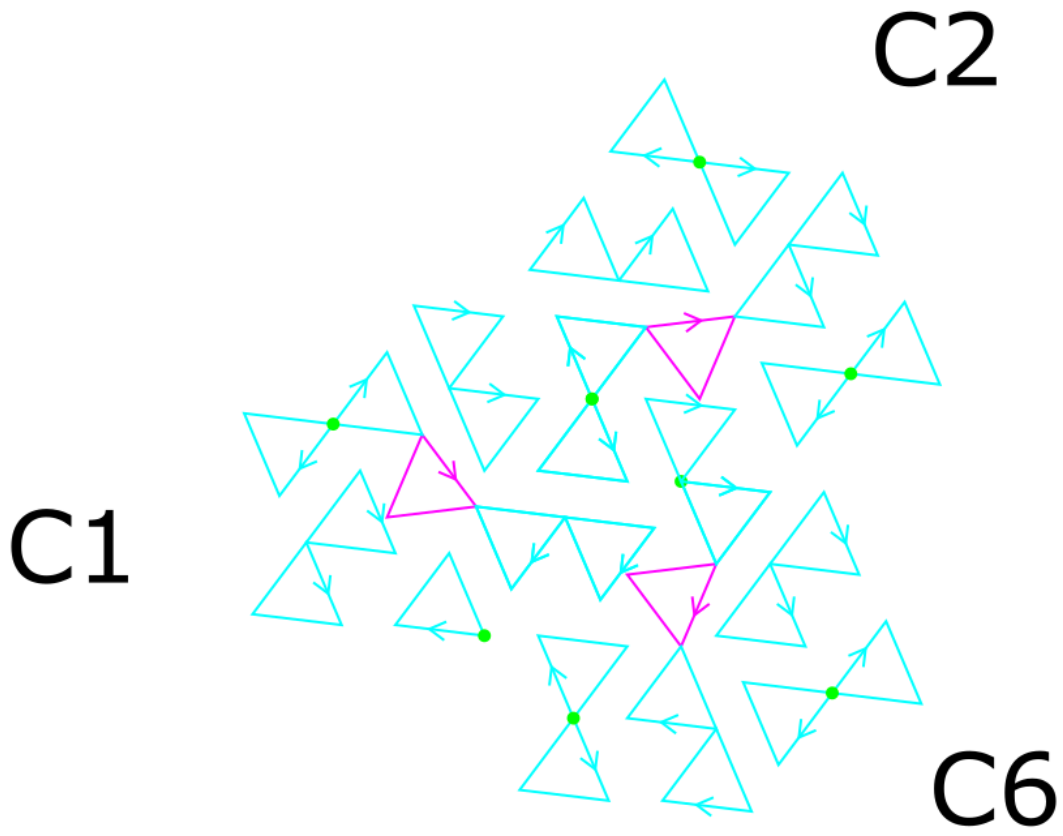




- Starting point for worms ☺
- After removing two half-turned tiles

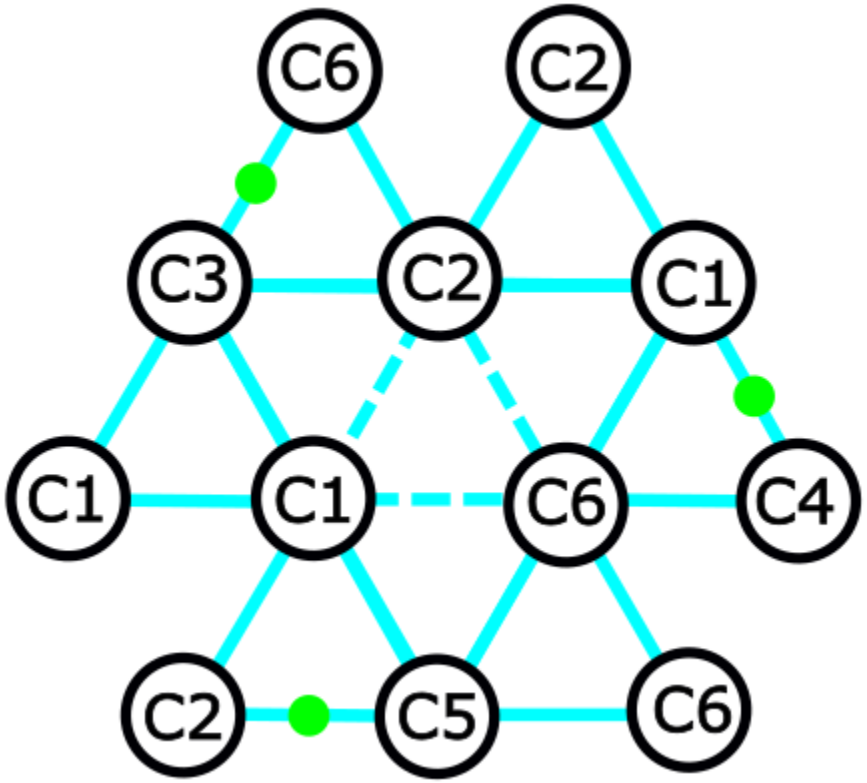
Defect healing





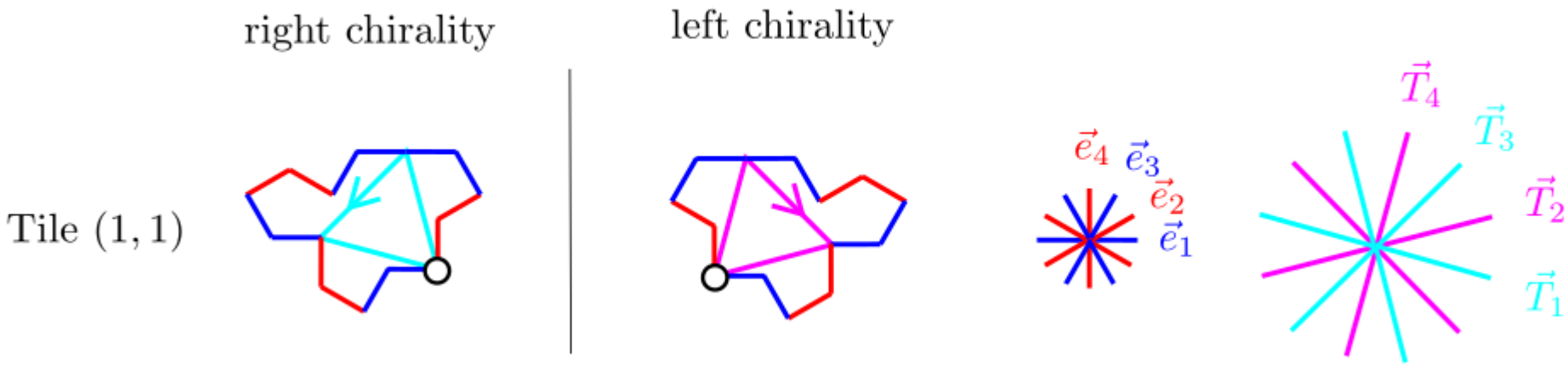
- Starting point for worms ☺
- After removing two half-turned tiles

TO DO: Step 2



- Replace all the 12 full clusters in this pattern with the same rule used previously for C1 (just a rotation of the pattern for C2, C3, C4, C5 and C6)

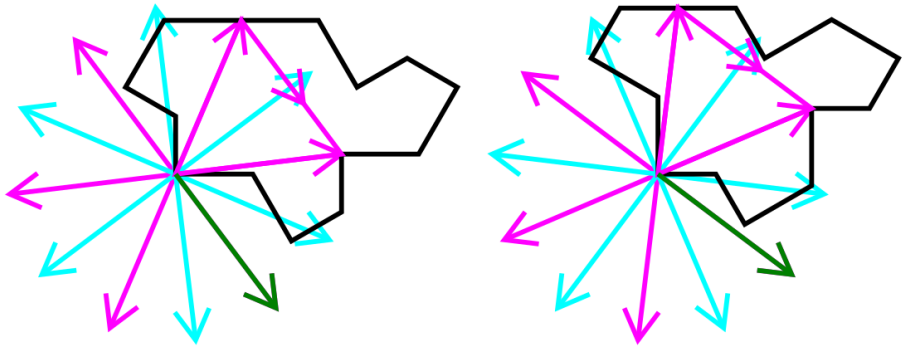
New reference triangle in Tile(a,b)



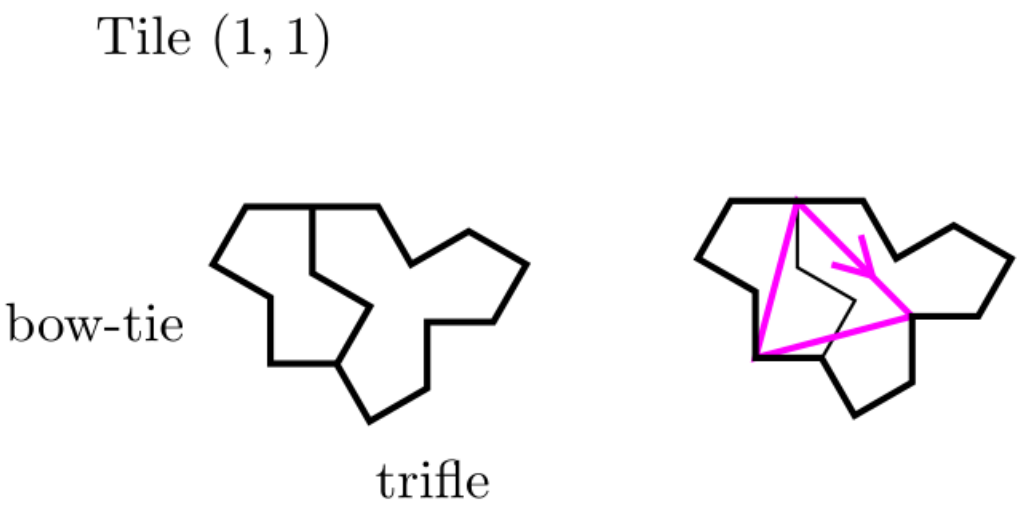
Robust for any T(a,b) tile ☺

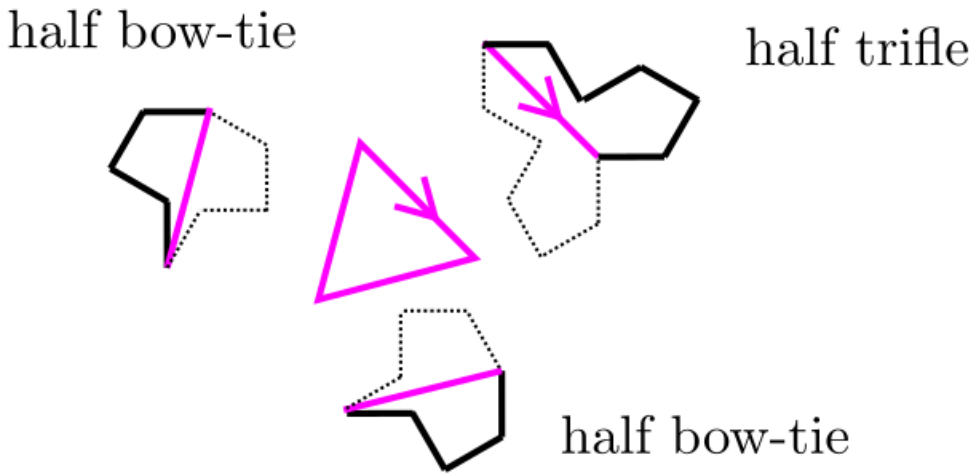
Tile(1.6,1.2) Tile(1.2,1.6)

a=1.6
b=1.2



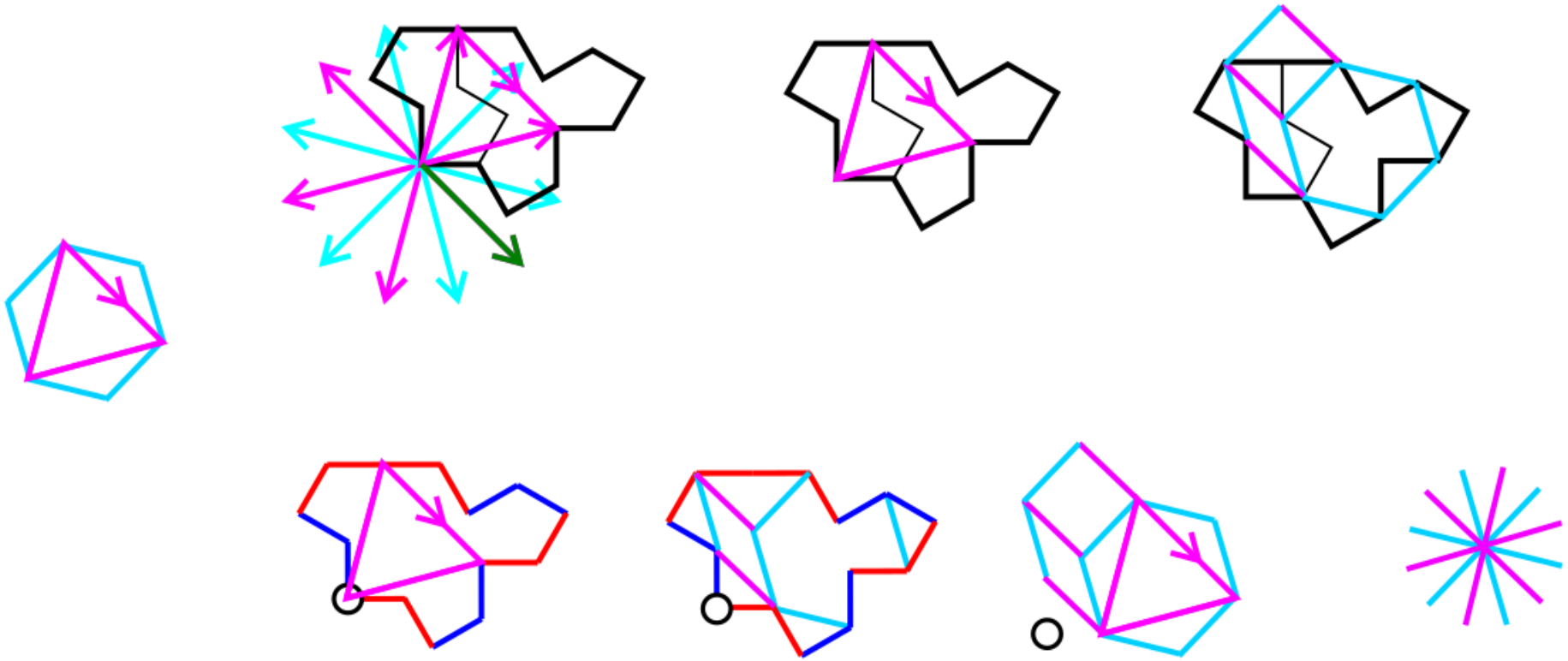
Link with trifle and bow-tie in Tile(1,1)





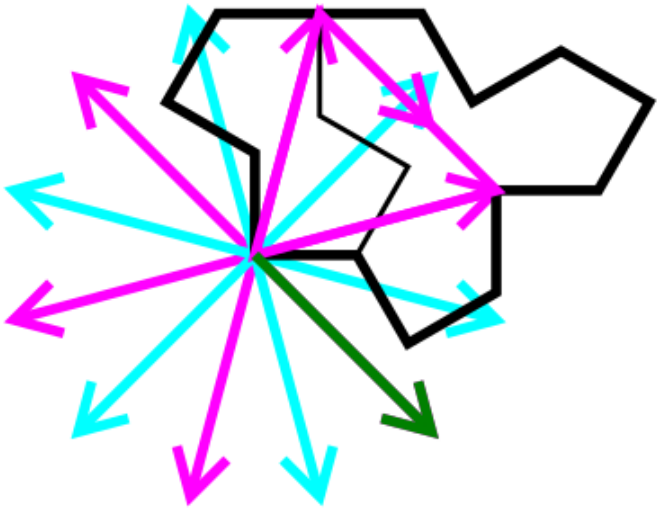
Comparison with hexagon/square/rhombus decomposition

Tile(1.0,1.0)

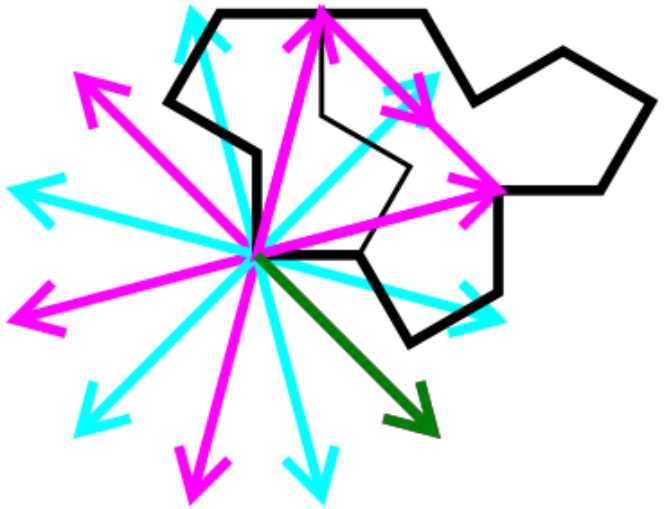


Tile(1,1) : spectre tile

Tile(1.0,1.0)

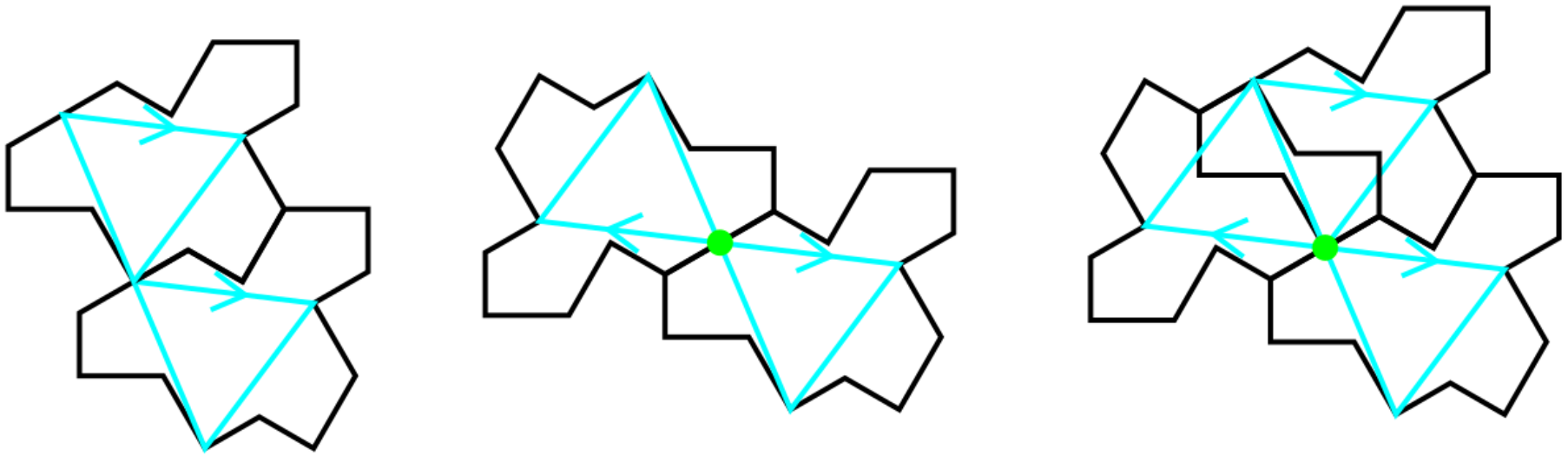


Tile(1.0,1.0)



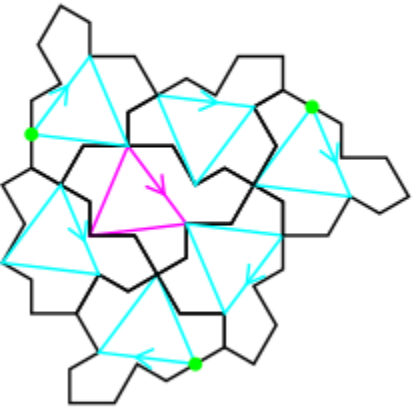
translation versus half-turn for any Tile(a,b)

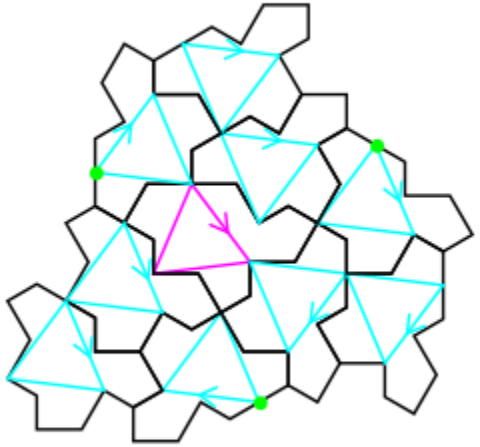
a=1.6
b=1.2

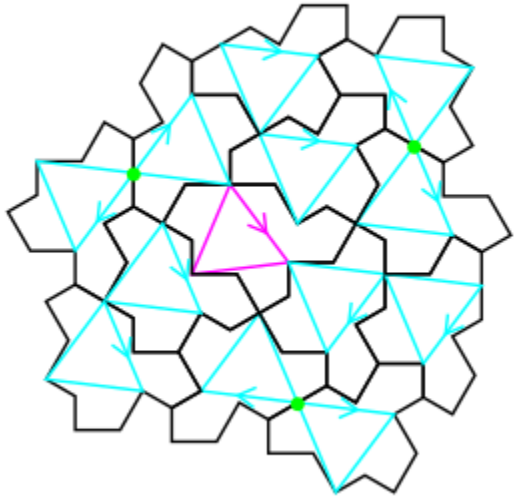


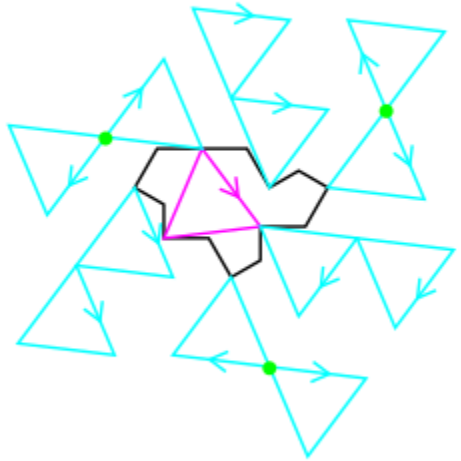
Cluster and full cluster

a=1.6
b=1.2







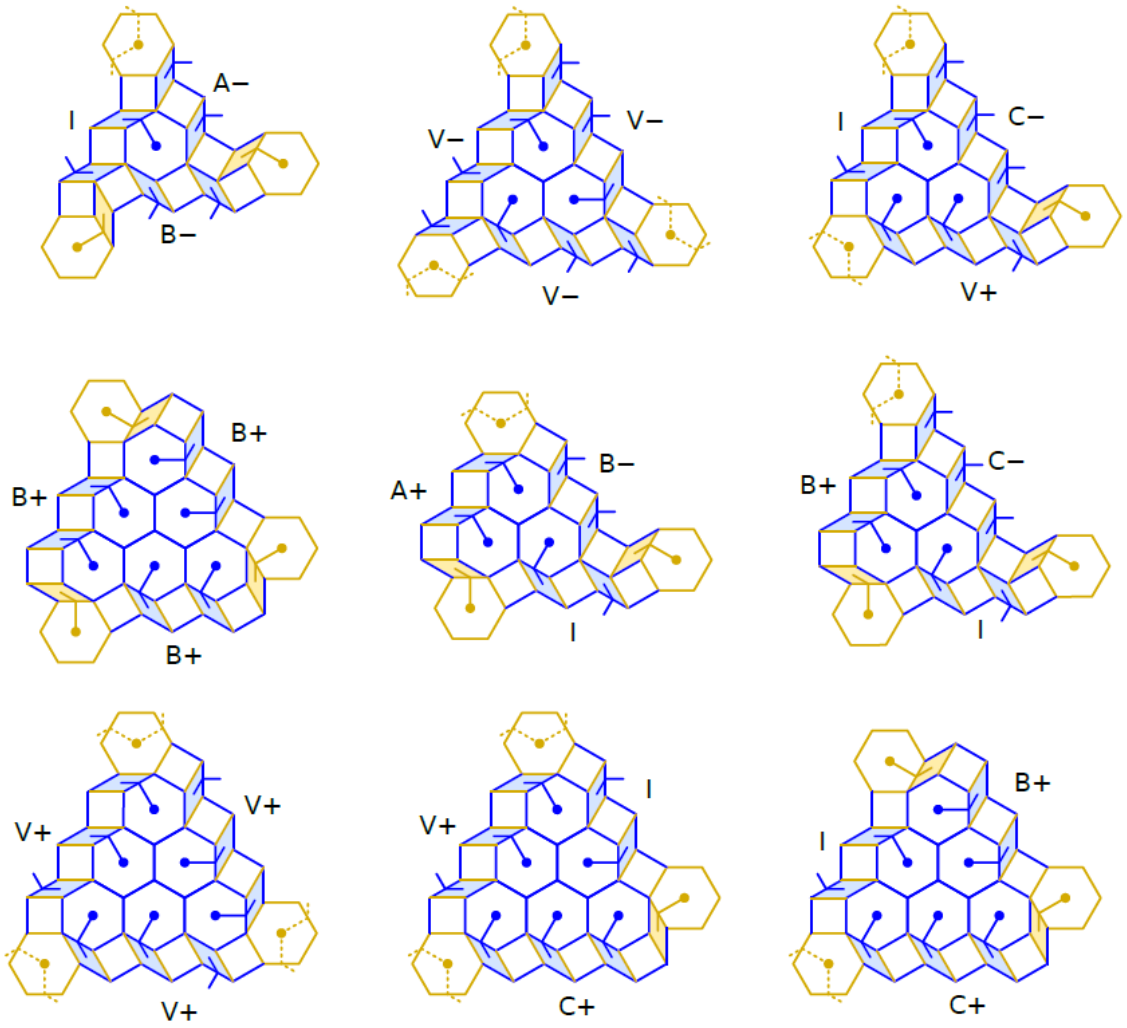


C cluster
(corona around a
ODD tile)

The three
translated tiles
are all in the tiling

Three more possible tiles
with half-turn around the
green points

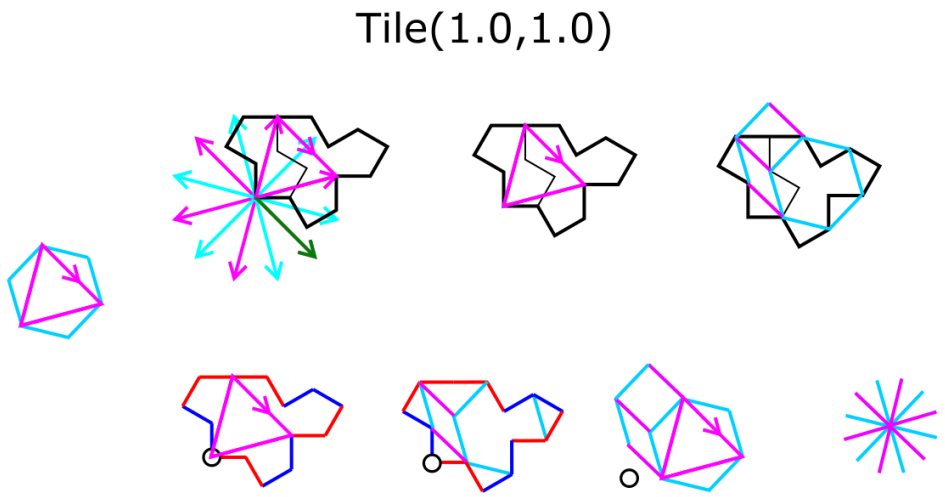
8 types of hexagonal clusters

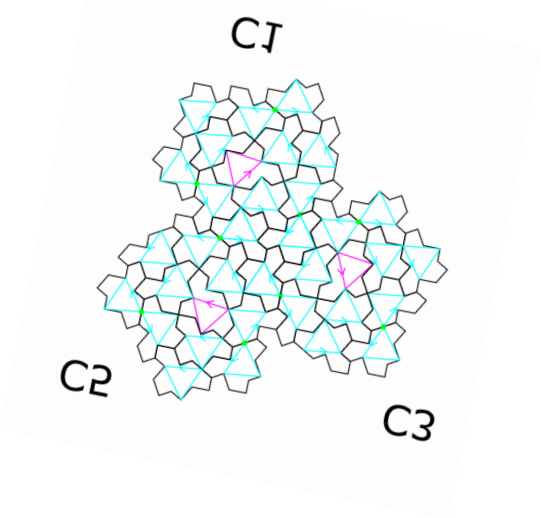


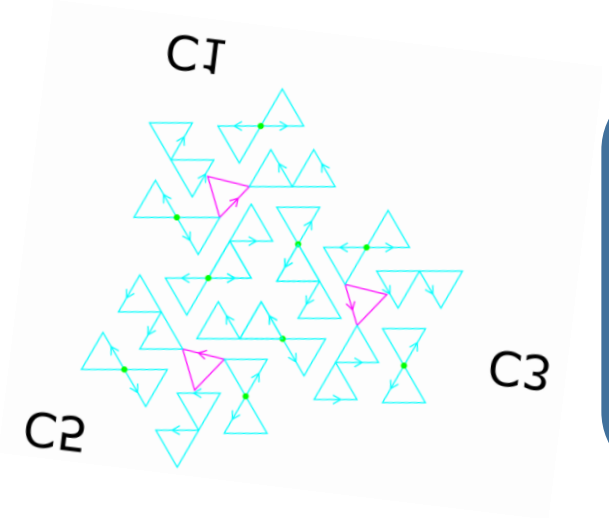
Combinatorial properties

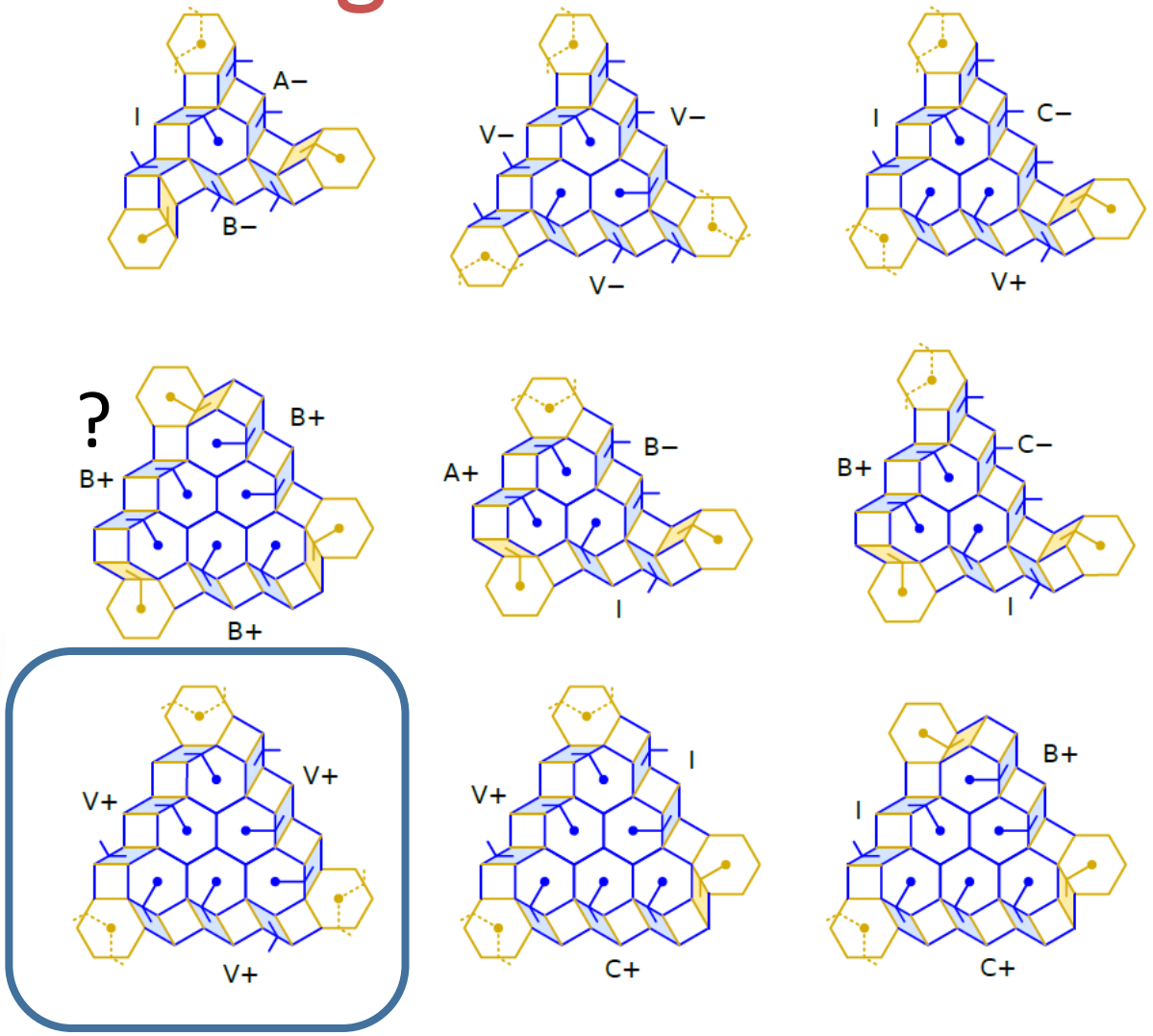
Chéritat, A., (2024). Observations on the hex clusters of the spectre tilings. https://arxiv.org/abs/2407.05359

Find equivalence with the hexagonal clusters ?



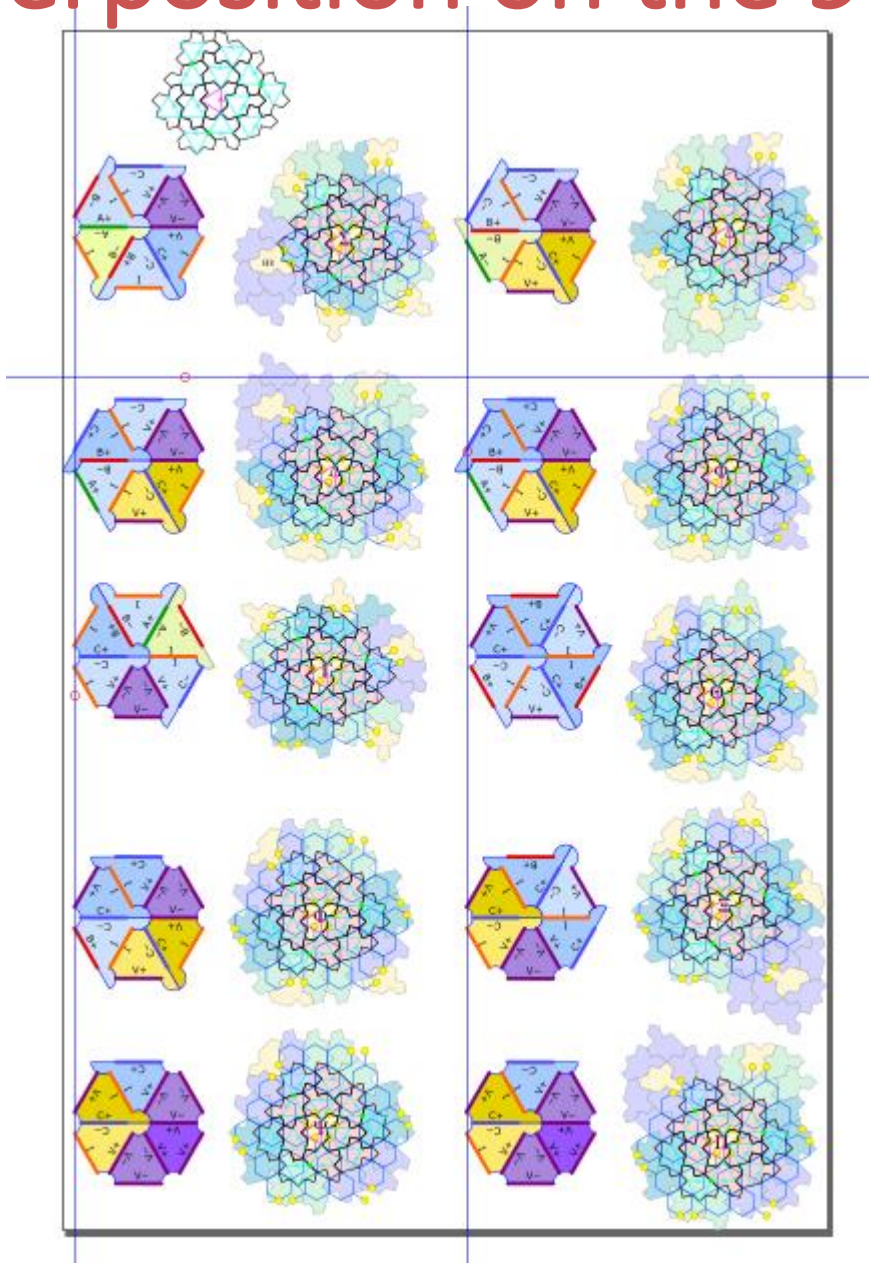






Chéritat, A., (2024). Observations on the hex clusters of the spectre tilings. https://arxiv.org/abs/2407.05359

Full cluster superposition on the 9 combinations



WHY ???

Homochiral python code on github

- Under construction !
- https://github.com/marimperorclerc/homochiral

python code for drawing clusters with Tile(a,b) in svg graphics.

References

Code is based on the drawsvg package:
https://pypi.org/project/drawsvg/
And the python code with alternate chirality for inflation for spectre tiling:
https://github.com/shrx/spectre/blob/master/spectre.py